

Three-Tier Architecture with EC2 (NGINX) + EKS (React/Node.js), Cross-Region RDS & DR –Single Stack Terraform Deployment

Work Effort Name	Three-Tier Architecture with EC2 (NGINX) + EKS (React/Node.js), Cross-Region RDS & DR –Single Stack Terraform Deployment
Prepared By	SOUNDARIYA B
Document Issued Datefot	06-08-2025
Document Version	3.0
Reviewed By	Mahavishnu Govindaraj
SA Certificate Number	

Version No.	Date	Revised by	Description of Changes
1	7/8/25	Immanuvel David Raj	Update cost specifications for all resources - updated Provide accurate resource configurations and security group details; remove any irrelevant information - updated Include a detailed explanation of RDS Multi-AZ configurations. Attach a complete and accurate flow diagram. Describe automation processes implemented using Ansible. Review IAM users, associated policies, and availability requirements - changed Clearly mention backup strategies and retention policies - updated Explain the monitoring strategy and tools in use – added cloudwatch and fluent bit.
2	11/8/25	Immanuvel David Raj	Update the "Environment which are being Migrated & Build:" remove unwanted rows. Alignments are incorrect in 1.2. Rework on the Infra setup Under capacity planning you have provided the AS-IS values but in the total core comparison you have provided NA. Rework on this. Update the security requirement Changes to be done in the TCO. Explain this RDS for database is suggested, since it will have features like automatic backups, automated software patching, and encryption at rest

Contents

1. Background & Introduction	4
1.1 Project Background.....	4
1.2 Project Scope	4
1.3 Solution Overview.....	5
1.4 Infra Setup	6
1.5 Capacity Planning.....	6
1.6 Availability Requirement	7
1.7 Security Requirement	8
1.8 Cost Details	8
1.9 Backup and Retention.....	9
1.10 Monitoring Tool Details	10
1.11 Patch Management	11
1.12 Contacts	11
1.13 Abbreviation	11
2. AWS Account	12
2.1 IAM Users & Password Policy	12
3. Architecture Diagrams	12
3.1 Common Service Network Architecture	12
3.2 Infra Design Diagram	13
3.3 CI/CD Pipeline	17
3.4 Triggering Pipeline	18
3.5 Design Consideration	19
4. AWS Account	19
4.1 Prod Route Table	20
4.2 Security Group Details	22

1. Background & Introduction

1.1 Project Background

Customer Background

A small finance company focused on building a scalable e-commerce platform to support MSME and SME businesses. Their mission is to empower small business owner especially women and underserved communities by providing digital tools that drive economic growth and community transformation.

Project Requirement

The project implements Cross-Region RDS Replication with a Disaster Recovery (DR) setup to ensure availability and data protection. An RDS read replica in a secondary region supports failover with Route 53 and Lambda automation, while web and app tiers run on EKS clusters across regions. Automated hourly backups with cross-region copy further strengthen recovery and business continuity.

Environment which are being Build:

Environments	Application
Production	Web-Application

1.2 Project Scope

The scope of the project comprises of the below Servers.

Application workloads	Prod
Web-App	4
Bastion Server	2
Database-RDS	1
Infra-server	1

All resources are provisioned in a secure and highly available VPC design leveraging public and private subnets, with multi-AZ deployments for EKS services and RDS, and security groups controlling access as per documented architecture.

- Deployment of web and application tiers on EKS clusters in both primary and DR regions.
- RDS cross-region replication with automated failover using Route 53 and Lambda.
- Automated hourly backups with cross-region copy for data protection.

- Configuration of monitoring, logging, and security controls.
- Regular DR failover testing to validate recovery readiness.

1.3 Solution Overview

Cross Region Replication approach:

- All application and infrastructure code is version-controlled in Git and managed by Jenkins pipelines to automate building, testing, and deploying across both primary (us-east-1) and DR (ap-south-1) AWS regions.
- Jenkins pipelines coordinate Docker image builds for the React frontend and Node.js backend, store images in Amazon ECR, and deploy to EKS clusters using Helm.
- Terraform automates provisioning of AWS resources: VPCs with non-overlapping CIDRs in both regions, public and private subnets in multiple availability zones, EKS clusters, EC2 (NGINX), RDS (with cross-region read replica), and Application Load Balancers.
- Route53 DNS with failover policy, plus VPC peering or Transit Gateway, provide resilient cross-region connectivity and enable automated failover and failback.
- Application Load Balancer distributes incoming traffic to workloads on EKS and NGINX servers running on EC2, which are configured automatically by Ansible.
- Sensitive credentials and secrets are securely managed with AWS Secrets Manager, with encryption provided by AWS KMS.
- Amazon RDS is configured for high availability in the primary region, with a cross-region read replica in the DR region for rapid failover; regular backups and automated DR drills are performed.
- CloudWatch and Fluent Bit enable centralized logging, monitoring, and automated alerting for infrastructure, applications, and DR events.
- IAM roles and security groups enforce least-privilege access and network segmentation, while encryption is enabled for all data at rest and in transit.
- IAM roles and security groups enforce least-privilege access and network segmentation, while encryption is enabled for all data at rest and in transit.
- All deployment and DR pipeline activities are logged for auditability and compliance, ensuring robust operational governance and reliability.

Disaster Recovery Approach:

- Warm Standby: Scaled-down EKS, EC2, and RDS replica provisioned in DR region (ap-south-1).
- Route 53 Failover: Health checks auto-redirect traffic to DR ALB on primary failure.
- Lambda Automation: Promotes RDS replica to primary during failover.
- Backups: Hourly RDS snapshots with cross-region copy and retention policy.
- Monitoring: CloudWatch, Fluent Bit, and SNS enable real-time alerts.
- Security: IAM, SGs, and KMS encryption enforced across both regions.
- DR Drills: Regular failover simulations ensure RTO/RPO compliance.

1.4 Infra Setup

Provision isolated VPCs with non-overlapping CIDRs in both us-east-1 (primary) and ap-south-1 (DR) regions, including public and private subnets with Internet and NAT Gateways for secure connectivity. Deploy EKS clusters, EC2 instances running NGINX, Application Load Balancers, and RDS MySQL with cross-region read replicas in both regions.

Automate infrastructure and app deployment using Terraform, Docker, Helm, and Jenkins CI/CD pipelines. Centralize monitoring with CloudWatch and Fluent Bit. Enforce strong security, enable AWS Client VPN for secure access and ongoing DR validation, and configure Route53 for automated DNS failover to ensure business continuity.

1.5 Capacity Planning

Initial Sizing Recommendations

Below is the list of Application names from (AS-IS) to (TO-BE-STATE)

Server Name	Environ ment	TO-BE-STATE		Instance Type	OS Version
		Core	RAM		
CR-Prod-us-east-App	Prod	2	4	t3.medium	Amazon Linux 2
CR-Prod-ap-south-App	Prod	2	4	t3.medium	Amazon Linux 2
CR-Prod-us-east-Nginx1	Prod	2	4	t3.medium	Amazon Linux 2
CR-Prod-us-east-Nginx2	Prod	2	4	t3.medium	
CR-Prod-ap-south-Nginx3	Prod	2	4	t3.medium	Amazon Linux 2
CR-Prod-ap-south-Nginx4	Prod	2	4	t3.medium	Amazon Linux 2

CR-Prod-us-east-RDS	Prod	2	4	db.t3.medium	Mysql (AWS)
CR-Prod-ap-south-RDS	Prod	2	4	db.t3.medium	Mysql (AWS)
CR-Bastion-server-us-east	Prod	2	4	t3.medium	Amazon Linux 2
CR-Bastion-server-ap-south	Prod	2	4	t3.medium	Amazon Linux 2

Rightsizing Instances

Instance Sizing will depend on the following aspects based on both performance as well as the cost:

- As-Is server sizing (CPUs, RAM & Storage)
- As-Is server utilization (minimum past 3 months max and avg utilization)
- As-Is server workload type (Web or Application / Database / Cache layer)
- As-Is server has specific high network throughput or IO throughput / IOPS or CPU intensive or Memory intensive.
- Type of Environment (Production / Non-Production)
- Network Bandwidth requirement

Total Core and Ram comparison:

Description	On-Prem	AWS
Core	20	20
RAM	40	40

1.6 Availability Requirement

To ensure high availability and disaster recovery, the solution leverages AWS multi-region architecture with active primary (us-east-1) and standby DR (ap-south-1) regions, supported by automated failover mechanisms and health checks.

- All critical resources—including EKS clusters, EC2 (NGINX), and RDS MySQL—are provisioned in multiple Availability Zones within each region to provide resilience against single-AZ failure.
- CloudWatch and Route53 health checks monitor service health, automatically triggering failover to the DR region if the primary becomes unavailable.
- EC2 Auto Recovery is enabled, and Route53 failover routing seamlessly directs traffic to healthy endpoints during an outage.

Note: All critical infrastructure and application resources are monitored and managed for auto-recovery, rapid failover, and failback. Regular DR drills and continuous health checks ensure readiness for unplanned outages and maintain a high service availability target of 99.9% or greater.

1.7 Security Requirement

S.NO	Layer	AWS
1	Host	I have leveraged Security Groups
2	Application	Implementing Application load balancer & make sure Application sites are encrypted with HTTPS/SSL
3	Network	Least-privilege Security Groups, NACLs for subnet-level control private subnets for application and DB with NAT Gateway attached.
4	Storage	KMS Encryption has applied In all EBS volumes and S3
	Database	KMS Encryption has leveraged for the DB Storage.
		SSL/TLS certificates has been leveraged in the DB secure Communication
		Credentials managed in AWS Secrets Manager.
5	Monitoring	Amazon CloudWatch for real-time monitoring and alerting
6	Other	Secrets Manager, KMS, IAM roles has been leveraged

1.8 Cost Details

Services	Region	Count
EC2 (t3.medium)	us-east-1	5
EC2(t3.medium)	ap-south-1	5
EC2(t3.large)	us-east-1	1
EKS	us-east-1	1
EKS	ap-south-1	1
RDS	us-east-1	1
RDS	ap-south-1	1
VPC	us-east-1	1
VPC	ap-south-1	1
SecretsManager	us-east-1	1
SecretsManager	ap-south-1	1
EBS	us-east-1	1
EBS	ap-south-1	1
S3	us-east-1	1

S3	ap-south-1	1
ECR	us-east-1	1

<https://calculator.aws/#/estimate?id=6b09222cf1ed87e9c838dc849c7d783910e537b9>

1.9 Backup and Retention

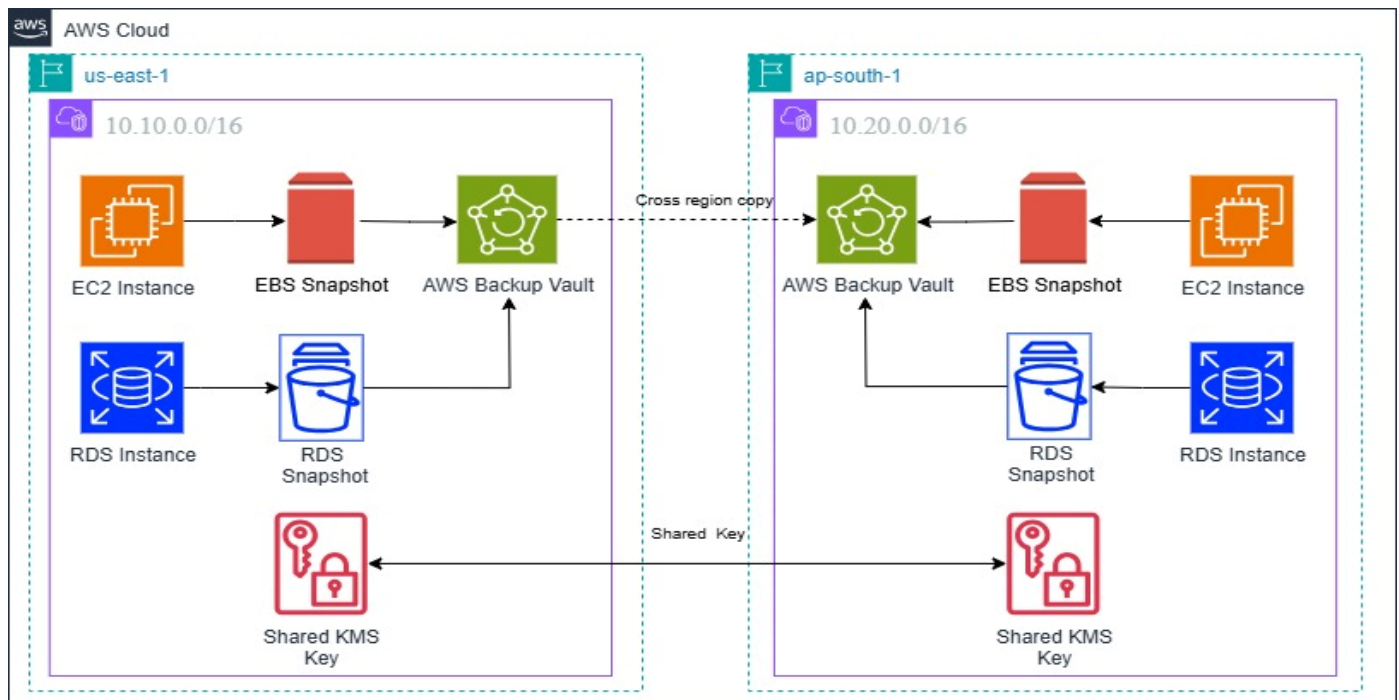
Automated hourly backups are taken for all critical AWS resources, including RDS database and EC2/EKS snapshots, in the primary region (us-east-1) and securely copied to the DR region (ap-south-1).

All backups are retained for 30 days to meet compliance and recovery needs.

Backup jobs are actively monitored, with alerts for any failures, and regular DR restore tests are performed to ensure data integrity and rapid recovery.

Environment	Server Name	Application Name	(AS-IS)Backup Frequency	(AS-IS) Type	(AS-IS) Backup Retention	(To-Be)Backup Frequency	(To-Be)Backup Retention	(To-Be) Region
Prod	CR-Prod-us-east-app	Web-app	Daily/Monthly	Comm Vault	60 Days	Hourly	30 Days	ap-south-1
Prod	CR-Prod-ap-south-app	Web-app	Daily/Monthly	Comm Vault	60 Days	Hourly	30 Days	us-east-1
Prod	CR-Prod-ap-south-Nginx	Nginx-server	Daily/Monthly	Comm Vault	60 Days	Hourly	30 Days	us-east-1
Prod	CR-Prod-us-east-Nginx	Nginx-server	Daily/Monthly	Comm Vault	60 Days	Hourly	30 Days	ap-south-1
Prod	CR-Prod-us-east-DB	DB-RDS	Daily/Monthly	Comm Vault	60 Days	Hourly	30 Days	ap-south-1
Prod	CR-Prod-ap-south-DB	DB-RDS	Daily/Monthly	Comm Vault	60 Days	Hourly	30 Days	us-east-1

Backup and DR Architecture:



1.10 Monitoring Tool Details

To ensure robust observability for all tiers of the architecture, configure AWS CloudWatch across EC2, EKS, and RDS components in both primary and DR regions. Use Fluent Bit as the log forwarder to collect, process, and ship logs from EKS workloads and EC2 instances to CloudWatch Logs for centralized visualization, alerting, and retention,

EC2 and EKS node metrics:

- CPUUtilization
- MemoryUtilization (via CloudWatch Agent)
- DiskSpaceUtilization
- NetworkIn / NetworkOut
- StatusCheckFailed

RDS metrics:

- CPUUtilization
- FreeStorageSpace
- DatabaseConnections
- BackupRetentionCheckFailed
- *ReplicaLag* (for cross-region DR readiness)

Note: By default, the threshold values for the above metrics are set to warning 80% and critical 90%.

Application Logs:

The CloudWatch agent is installed on all application servers (EC2 and containers) to stream detailed application and system logs into predefined CloudWatch Log Groups.

AWS Services:

- EC2 Metrics
- Load balancer metrics
- Node Group metrics
- RDS Metrics

1.11 Patch Management

Security: Patch management fixes vulnerabilities on your software and applications that are susceptible to cyber-attacks, helping your organization reduce its security risk.

System uptime: Patch management ensures your software and applications are kept up-to-date and run smoothly, supporting system uptime.

Compliance: With the continued rise in cyber-attacks, organizations are often required by regulatory bodies to maintain a certain level of compliance. Patch management is a necessary piece of adhering to compliance standards.

Feature improvements: Patch management can go beyond software bug fixes to also include feature/functionality updates. Patches can be critical to ensuring that you have the latest and greatest that a product has to offer.

Patch Window:

Environment	Frequency	Feature
Production	6 months	Security

1.12 Contacts

Owner	Role	Name	Email ID
1CH	Sr.Solution Architect	Mahavishnu Govindaraj	vishnu@1cloudhub.com
1CH	Solution Architect	Immanuvel David Raj	immanuvel@1cloudhub.com

1.13 Abbreviation

The following are the terminology and acronyms used in the document.

S.NO	Abbreviation / Keyword	Definition
1	AWS	Amazon Web Services
2	EC2	Elastic Compute Cloud
3	VPC	Virtual Private Cloud
4	KMS	Key Management Service
5	ALB	Application Load Balancer

6	NAT	Network Address Translation
7	EKS	Elastic Kubernetes Service
8	EBS	Elastic Block Storage
9	IAM	Identity and Access Manager
10	RDS	Relational Database Service

2. AWS Account

Below are the AWS accounts for Prod & Sandbox environment setup.

S.NO	Account Number	Account Name	Description & Purpose
1			
2			
3			

2.1 IAM Users & Password Policy

Below are the IAM user accounts to be created for 1CloudHub Team.

2.1.1 IAM Users

1CH Team

AWS Account Name	Username	Policies
IAM-user	1cloudhub-infra	Administrator Access

2.1.2 Password Policy

Password policy for AWS accounts

Password policy Info
Edit

Configure the password requirements for the IAM users.

This AWS account uses the following default password policy:

Password minimum length
8 characters

Password strength
Include a minimum of three of the following mix of character types:

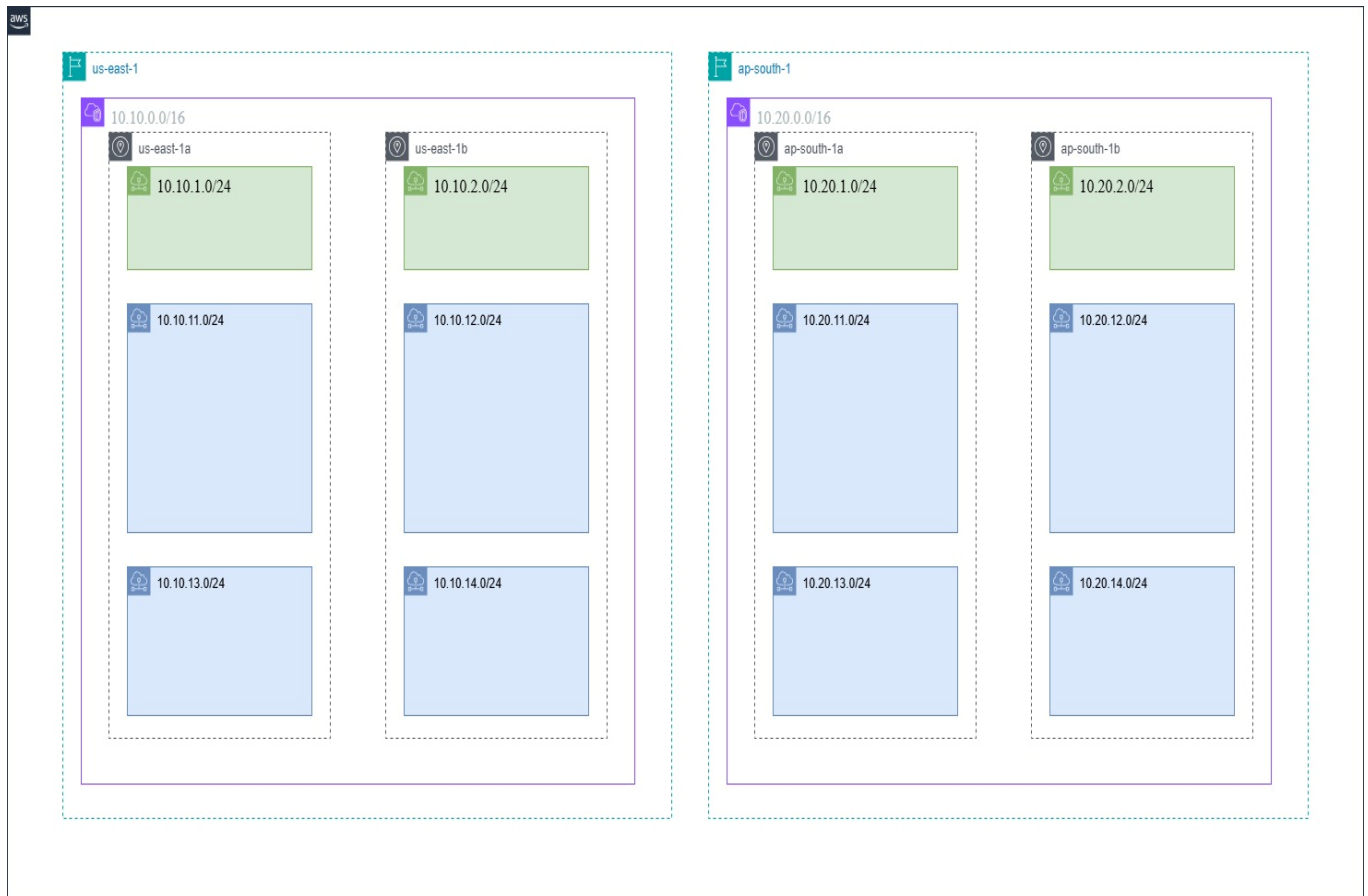
- Uppercase
- Lowercase
- Numbers
- Non-alphanumeric characters (!@#\$%^&*()_+-=[]{}|')

Other requirements

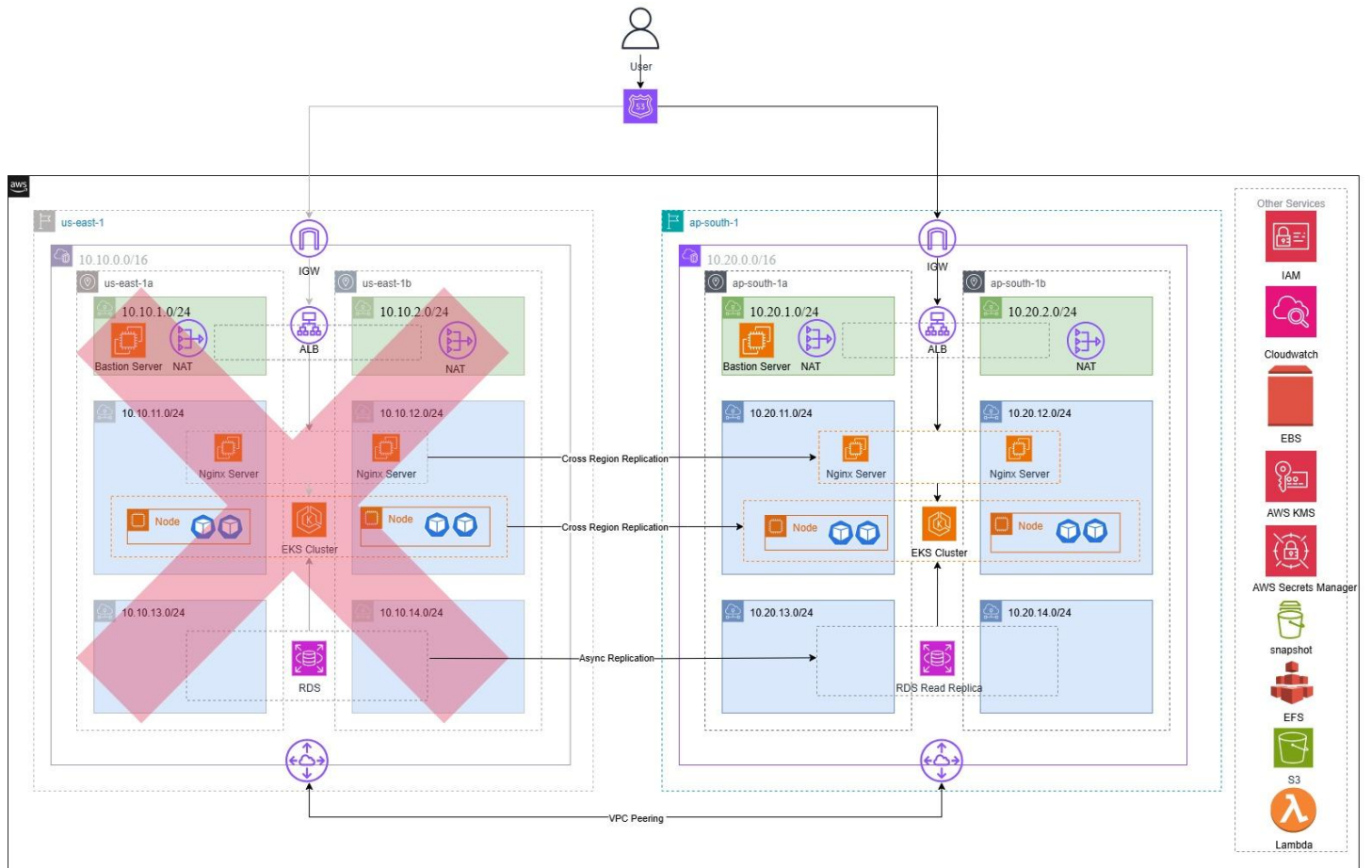
- Never expire password
- Must not be identical to your AWS account name or email address

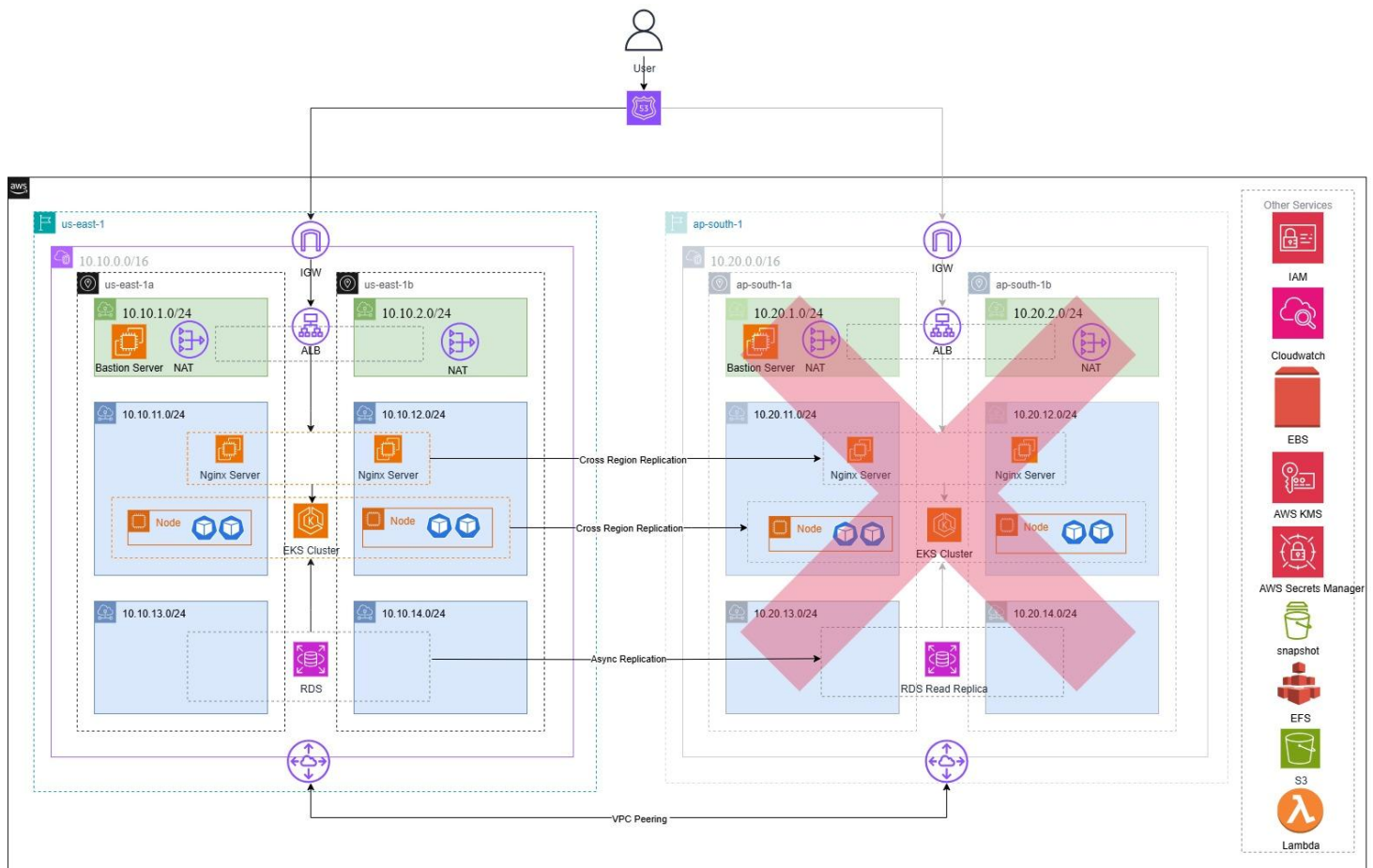
3. Architecture Diagrams

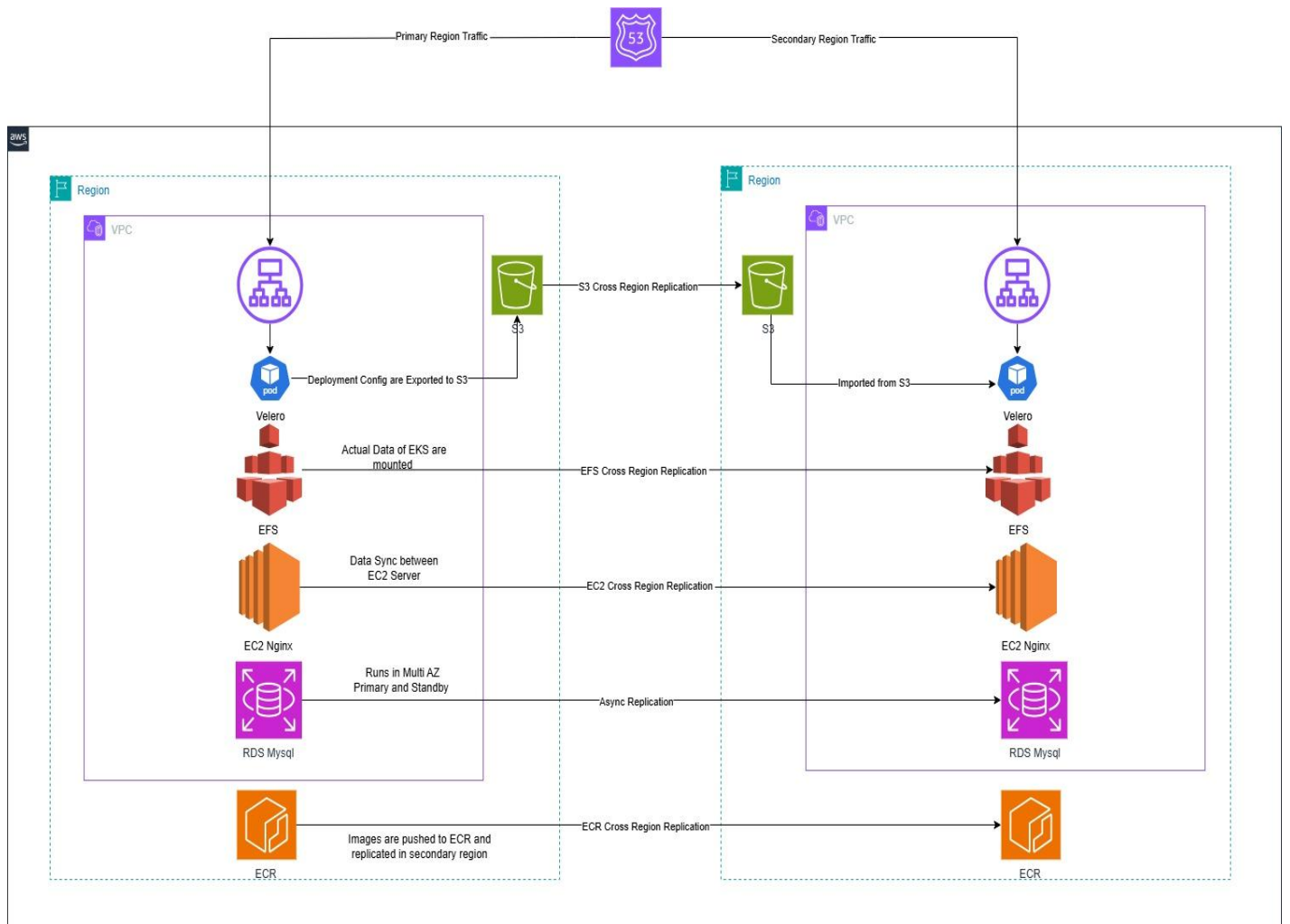
3.1 Common Service Network Architecture



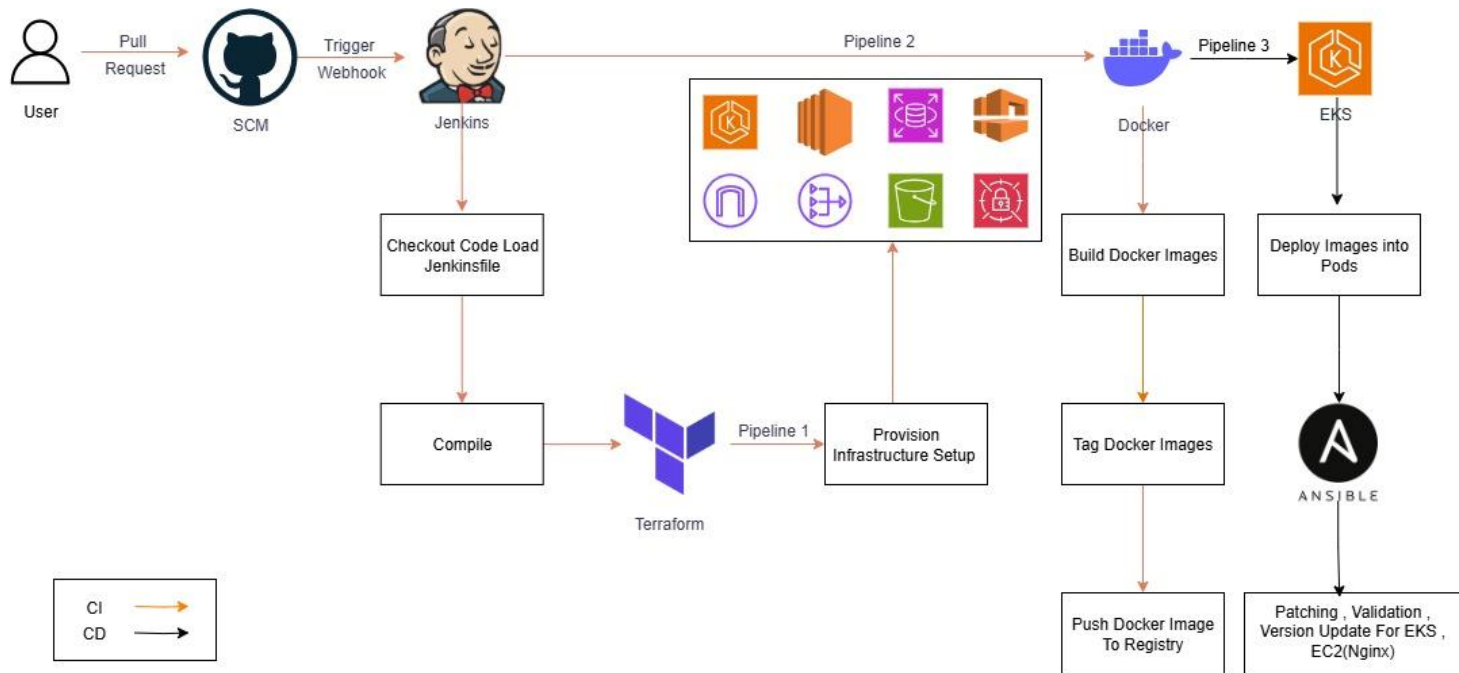
3.2 Infra Design Diagram



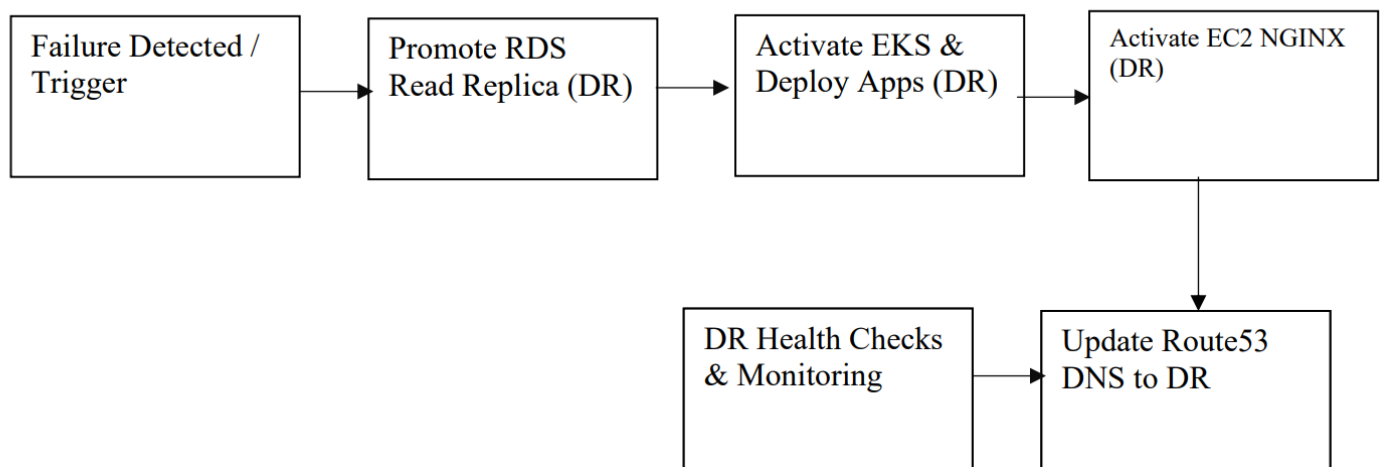




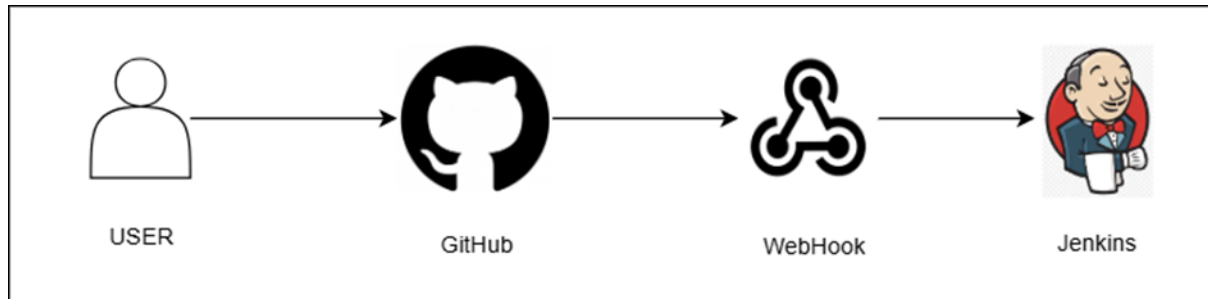
3.3 CI/CD Pipeline



DR Pipeline



3.4 Triggering Pipeline



3.5 Design Consideration

Design Consideration for Prod Common service environment.

Category	Component	Consideration	Services Used
Reliability	High Availability	Application been deployed across multiple AZs in both regions. (ECS Cluster) and RDS DB have	ALB is leveraged, Multi-AZ ECS Service, Multi-AZ RDS (with cross-region read replica as required)
Performance Measures	Scaling	Vertical and horizontal scaling can be done for App / DB servers, (Downtime required)	No autoscaling required / Manual Scaling
Security	Network	EC2 and DB servers will be launched in private subnet.	NAT gateway has leveraged
	Application	Frontend and Backend are running as pods in cluster exposed via ALB	Cluster with ALB has leveraged
	Host Access	All EC2 instance access are restricted at Security Group.	The application Ec2 instances having only the terminal server IP are whitelisted in the SG
	User Access	User access will be managed by 1CH team	IAM with MFA has leveraged
	AWS Portal	Access to AWS resources is Controlled by IAM user policy with organizational power user access enabled.	For Console users are created in the account with MFA.
	Data at Rest	For EBS encryption KMS has used in all accounts	Custom KMS has leveraged to encrypt the data
Ease of Management	Backup	Backup will be enabled for all prods including OS and Data disk.	AWS backup services enabled for EBS with retention policy for 30 days

4. AWS Account

Virtual private cloud (VPC) is an on-demand configurable pool of shared computing resources allocated within a public cloud environment, providing a certain level of isolation between the different organizations using the resources.

VPC	Region	VPC CIDR	VPC ID
us-east-1-main vpc	us-east-1	10.10.0.0/16	TBD
ap-south-1-dr vpc	ap-south-1	10.20.0.0/16	TBD

Subnet is a logical partition of an IP network into multiple, smaller network segments and it is tightly linked to IP addresses, subnet masks, and CIDR notation.

Subnet Tier	Subnet Name	Subnet CIDR	Available IPs	Subnet ID
Public Subnet	Server-Subnet-pri-1A	10.10.1.0/24	256	TBD
Public Subnet	Server-Subnet-pri-1B	10.10.2.0/24	256	TBD
Private Subnet	app-Subnet-pri-1A	10.10.11.0/24	256	TBD
Private Subnet	app-Subnet-pri-1B	10.10.12.0/24	256	TBD
Private Subnet	DB-Subnet-pri-1A	10.10.13.0/24	256	TBD
Private Subnet	DB-Subnet-pri-1B	10.10.14.0/24	256	TBD
Public Subnet	Server-Subnet-DR-1A	10.20.1.0/24	256	TBD
Public Subnet	Server-Subnet-DR-1B	10.20.2.0/24	256	TBD
Private Subnet	app-Subnet-DR-1A	10.20.11.0/24	256	TBD
Private Subnet	app-Subnet-DR-1B	10.20.12.0/24	256	TBD
Private Subnet	DB-Subnet-DR-1A	10.20.13.0/24	256	TBD
Private Subnet	DB-Subnet-DR-1B	10.20.14.0/24	256	TBD

It is an optional layer of security for your VPC that acts as a firewall for controlling traffic in and out of one or more subnets.

Currently we are not blocking any incoming and outgoing traffic using NACL

A route table contains a set of rules, called routes, that are used to determine where network traffic is directed. Each subnet in your VPC must be associated with a route table.

4.1 Prod Route Table

us-east-1-main-vpc:

Route Table Name	Destination	Target	Subnet Association
us-east-1-main-vpc-Public-RT	10.10.0.0/16	local	Server-Subnet-pri-1A Server-Subnet-pri-1B
	0.0.0.0/0	IGW	

Route Table Name	Destination	Target	Subnet Association
us-east-1-main-vpc-Private-RT-1	10.10.0.0/16	local	app-Subnet-pri-1A
	0.0.0.0/0	nt-1a	

Route Table Name	Destination	Target	Subnet Association
us-east-1-main-vpc-Public-RT-2	10.10.0.0/16	local	App-Subnet-pri-1B
	0.0.0.0/0	nat-1b	

Route Table Name	Destination	Target	Subnet Association
us-east-1-main-vpc-DB-RT	10.10.0.0/16	local	DB-Subnet-pri-1A DB-Subnet-pri-1B

ap-south-1-DR-vpc:

Route Table Name	Destination	Target	Subnet Association
ap-south-1-main-vpc-Public-RT	10.20.0.0/16	local	Server-Subnet-DR-1A Server-Subnet-DR-1B
	0.0.0.0/0	IGW	

Route Table Name	Destination	Target	Subnet Association
ap-south-1-main-vpc-Private-RT-1	10.20.0.0/16	local	app-Subnet-DR-1A
	0.0.0.0/0	nt-1a	

Route Table Name	Destination	Target	Subnet Association
ap-south-1-main-vpc-Public-RT-2	10.20.0.0/16	local	app-Subnet-DR-1B
	0.0.0.0/0	nat-1b	

Route Table Name	Destination	Target	Subnet Association
ap-south-1-main-vpc-DB-RT	10.20.0.0/16	local	DB-Subnet-DR-1A DB-Subnet-DR-1B

EC2 Instance Security Group

Security groups are associated with EC2 instances and provide security at the protocol and port access level. Each security group act as the same way as a firewall and it contains a set of rules that filter traffic coming into and out of an EC2 instance.

4.2 Security Group Details

Security Group Name	Protocol	Ports	Source
Bastion SG	SSH	22	49.207.186.65
EKS-Worker-SG	TCP,SSH	443,22	10.10.13.0/24,10.10.14.0/24, 10.20.13.0/24,10.20.14.0/24
DB-SG	TCP	3306	10.10.11.0/24,10.10.12.0/24, 10.20.11.0/24,10.20.12.0/24
ALB-SG	HTTP, HTTPS	80, 443	0.0.0.0/0
Infra-SG	Custom TCP	8080,5985 (HTTP) or 5986 (HTTPS)	49.207.186.65

4.4.1 Tagging

Below tags will be used for the migration purpose

- 1) Environment
- 2) Application Name
- 3) Application Role

Example:

1. Environment = Production
2. Application Name = Web App
3. Application Role = APP

Enabled To track the event history of AWS account activity of any API activity in all OUs. Below are the S3 buckets for log management for Audit trail.

Note: Since CloudTrail provides only 90 days so log stored in bucket is recommended.

Prod S3 Bucket Name	Purpose	Retention (Days)
App-Storage	Cloud Trail	90

4.3 Config Details

AWS Config is a service that enables you to assess, audit, and evaluate the configurations of your AWS resources. Config continuously monitors and records your AWS resource configurations and allows you to automate the evaluation of recorded configurations against desired configurations.

Prod:

Name	Remediation action	Type	Compliance
encrypted volumes	Not set	AWS managed	-
cloudtrail-security-trail-enabled	Not set	AWS managed	-
cloud-trail-encryption-enabled	Not set	AWS managed	-
acm-certificate-expiration-check	Not set	AWS managed	-
db-instance-backup-enabled	Not set	AWS managed	-
account-part-of-organizations	Not set	AWS managed	-
guardduty-non-archived-findings	Not set	AWS managed	-
iam-policy-no-statements-with-admin-access	Not set	AWS managed	-
iam-password-policy	Not set	AWS managed	-
iam-root-access-key-check	Not set	AWS managed	-
mfa-enabled-for-iam-console-access	Not set	AWS managed	-

Name	Remediation action	Type	Compliance
ec2-stopped-instance	Not set	AWS managed	-
cloudtrail-enabled	Not set	AWS managed	-
multi-region-cloudtrail-enabled	Not set	AWS managed	-
guardduty-enabled-centralized	Not set	AWS managed	-
ec2-volume-inuse-check	Not set	AWS managed	-
alb-waf-enabled	Not set	AWS managed	-
cloud-trail-log-file-validation-enabled	Not set	AWS managed	-
alb-http-to-https-redirection-check	Not set	AWS managed	-
iam-user-mfa-enabled	Not set	AWS managed	-
s3-bucket-versioning-enabled	Not set	AWS managed	-
s3-bucket-public-read-prohibited	Not set	AWS managed	-
s3-bucket-server-side-encryption-enabled	Not set	AWS managed	-
vpc-flow-logs-enabled	Not set	AWS managed	-
s3-bucket-ssl-requests-only	Not set	AWS managed	-
s3-account-level-public-access-blocks	Not set	AWS managed	-
wafv2-logging-enabled	Not set	AWS managed	-
root-account-mfa-enabled	Not set	AWS managed	-
s3-bucket-level-public-access-prohibited	Not set	AWS managed	-
vpc-default-security-group-closed	Not set	AWS managed	-

Enabled To track the event history of AWS account activity of any API activity.

S.No	Cloud Trail status	Account details
1	Enabled	Prod

AWS Key Management Service (KMS) is an Amazon Web Services product that allows administrators to create, delete and control keys that encrypt data stored in AWS databases and products. AWS KMS provides a single view of all AWS keys in use, creating centralized encryption control.

Prod:

AWS Managed Keys

Aliases	Key ID	Status	Key spec
aws/acm	TBD	Enabled	SYMMETRIC_DEFAULT
aws/backup	TBD	Enabled	SYMMETRIC_DEFAULT
aws/ebs	TBD	Enabled	SYMMETRIC_DEFAULT

Customer Manager keys

Aliases	Key ID	Status	Key spec	Key usage
NA	NA	NA	NA	NA

AWS Managed Keys

Aliases	Key ID	Status	Key spec
aws/ebs	TBD	Enabled	SYMMETRIC_DEFAULT
aws/backup	TBD	Enabled	SYMMETRIC_DEFAULT
aws/acm	TBD	Enabled	SYMMETRIC_DEFAULT

Customer Manager keys

Aliases	Key ID	Status	Key spec	Key usage
NA	NA	NA	NA	NA

AWS Security HUB used to comprehensive view of your high-priority security alerts and compliance status across AWS accounts. With Security Hub, you now have a single place that aggregates, organizes, and prioritizes your security alerts, or findings, from multiple AWS services, such as Amazon Guard Duty, Amazon Inspector, and Amazon Macie, as well as from AWS Partner solutions.

With Security HUB we have enabled the CIS AWS foundational Benchmark and AWS Foundational Security has enabled in all accounts.